

AYUSH WAGHE

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EDUCATION

Computer Science and Engineering	Vellore Institute of Technology (Vellore)	2025	CGPA:- 8.7
Class-XII (Maharashtra State Board)	City Pride Junior College, Pune	2021	87.33%
Class-X (CBSE)	City Pride School, Pune	2019	95%

SKILLS

- Java, Spring Boot, Spring Security, Kafka, PostgreSQL, React, Docker, SQL, Git, HTML/CSS
- Machine Learning, Deep Learning

EXPERIENCE

- **Research Intern | Samsung Research Institute (Bangalore)** **August-December 2023**
 - Engineered a **TensorFlow-based** computer vision solution for detecting ghost artifacts in HDR images as part of **Samsung PRISM**.
 - Evaluated and optimized the detection pipeline to improve reliability across diverse HDR image datasets.
- **Software Development Engineer in Test (SDET) | Tata Consultancy Services (Mumbai)** **July 2025-Present**
 - Designed and enhanced a Java-based automation framework with a modular architecture following the **Page Object Model (POM)**, improving maintainability and reusability.
 - Optimized framework execution by implementing configurable batch-based Excel I/O (default batch size: 50), reducing repetitive file operations and decreasing overall test execution time by approximately **40%**.
 - Implemented a configurable retry mechanism for failed test cases with exception handling, improving framework resilience against transient failures and reducing unnecessary manual reruns.
 - Refactored the framework into reusable Action, Validation, and utility layers, simplifying new test development and reducing code duplication.
 - Improved framework reliability by implementing automatic Excel backup rotation and Excel file-lock detection to prevent corruption of excel test data.
 - Enhanced reporting by integrating Extent Reports with module-wise report generation and automatic screenshot capture for faster debugging and execution analysis.

PROJECTS

- **Savings Saga | Java full Stack (Spring Boot, React, PostgreSQL, Kafka, Docker)** | [\[Github\]](#) | [\[Live demo\]](#)
 - Built a full-stack personal finance management application using **Spring Boot, React, PostgreSQL, and Spring Security**.
 - Built secure **JWT-based authentication** with **HttpOnly cookies** and **Spring Security** for stateless session management.
 - Integrated **Apache Kafka** for asynchronous event processing to power bill reminder notifications and decouple business workflows.
 - Implemented **scheduled jobs** for automatic recurring bill generation and overdue bill processing.
 - Optimized **PostgreSQL query** performance by designing **indexes** for high-frequency transaction and analytics queries.

CERTIFICATIONS

- AWS Certified Cloud Practitioner

ACHIEVEMENTS

- Published a peer-reviewed research paper in Scientific Reports (Nature Portfolio) on machine learning techniques for banana plant health analysis. [\[Link\]](#)
- Final-year project recognized with patent publication selection by VIT for research on HDR image ghost artifact removal.